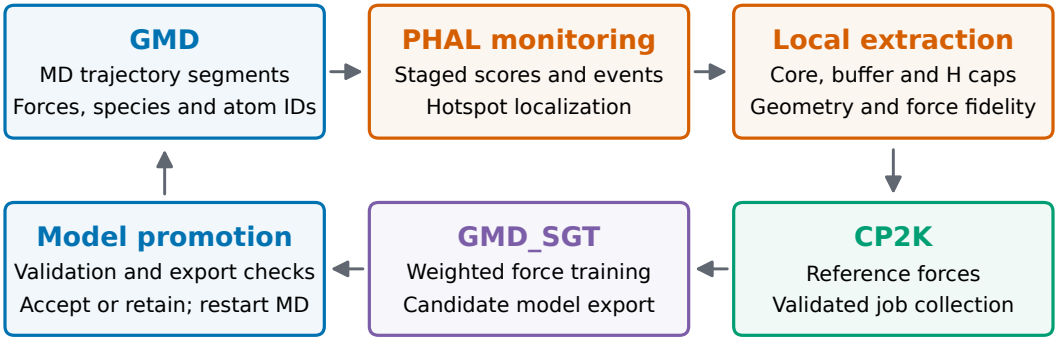A local reference-labeling loop in the GMD ecosystem